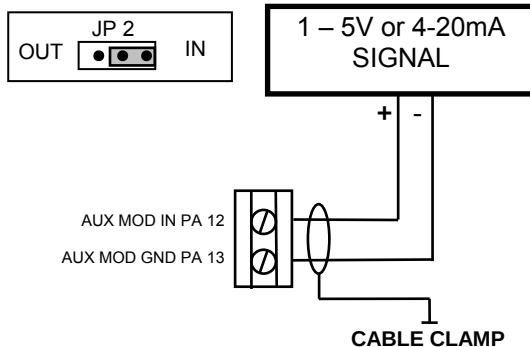




WARNING

Do not wire the Sensor “live” with power on. With JPI in the “IN” position a 220 ohm resistor is put across terminals PB9 and PB10. Shorting or miss-wiring these terminals may damage this resistor. Should the resistor become open, an external 220 ohm +/- 1% ½ watt resistor may be installed across terminals PB9 and PB10.

2.6.17 Remote Setpoint (Analog Input 5) (PPC6000)



The auxiliary modulation input cabling must be overall screened PVC insulated minimum 24 AWG (7/0.2mm) (number of cores as required by the relevant sensor). Since this cable may be run in conduit with high voltage wiring, it's voltage rating must exceed the maximum voltage carried by any other cable connected to the control or run in the same conduit. Ensure link JP2 is set to the 'IN' position if the input is being connected to a device with a 4-20mA output.

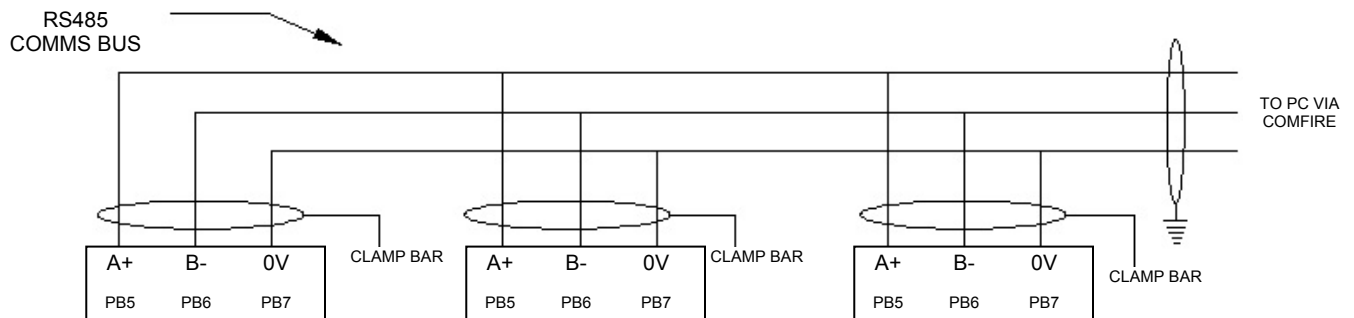
Incorrect connection may damage or destroy the units being connected.

2.6.18 RS485 interface (PPC6000)

The RS485 interface included in the basic control is intended to be used for interconnection of several Fireye units or connection of a 'lap-top' computer during commissioning if it is intended to make a permanent connection to a PC or non-Fireye equipment the optional isolated RS485 daughter board should be installed. The RS485 cabling **MUST** be overall braided shielded (screened) 3-core PVC insulated 24 AWG (7/0.2mm). Since this cable may be run adjacent to, and/or in the same conduit as high voltage wiring, its voltage rating must exceed the maximum voltage carried by any other cable connected to the control or run in the same conduit.

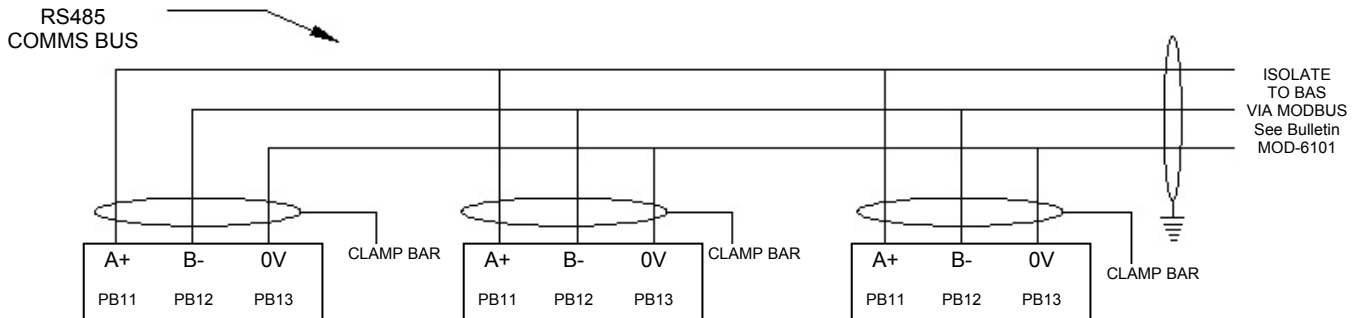
Terminate the screen **at each control** using the cable clamps provided. **Incorrect connection or application of excess voltage may damage or destroy the units being connected.**

Sequencing



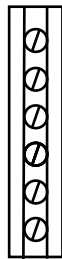


Building Automation



NOTE: Due to differences in communications hardware, it may be necessary to reverse the “A” and “B” wires.

RS485 A (+) PB 5
RS485 B (-) PB 6
RS 485 GND PB 7
RS 485 ISOL A (+) PB 11
RS485 ISOL B (-) PB 12
RS485 ISOL GND PB 13



For details on connecting PPC6000 series controls to a PC on the communications bus, refer to the manual NX-1502 (Comfire2 combustion analysis tool).

NOTE: Isolated RS485 (PB11 thru PB13) for use with Modbus / BAS integration – Requires optional Daughter Board. See MOD-6101 for details.

NOTE: Interposing terminal blocks should be avoided when shielded cable is required. Interposing terminals present a risk of electrical noise interference resulting in unreliable operation.

2.6.19 Variable Speed Drive Selection:

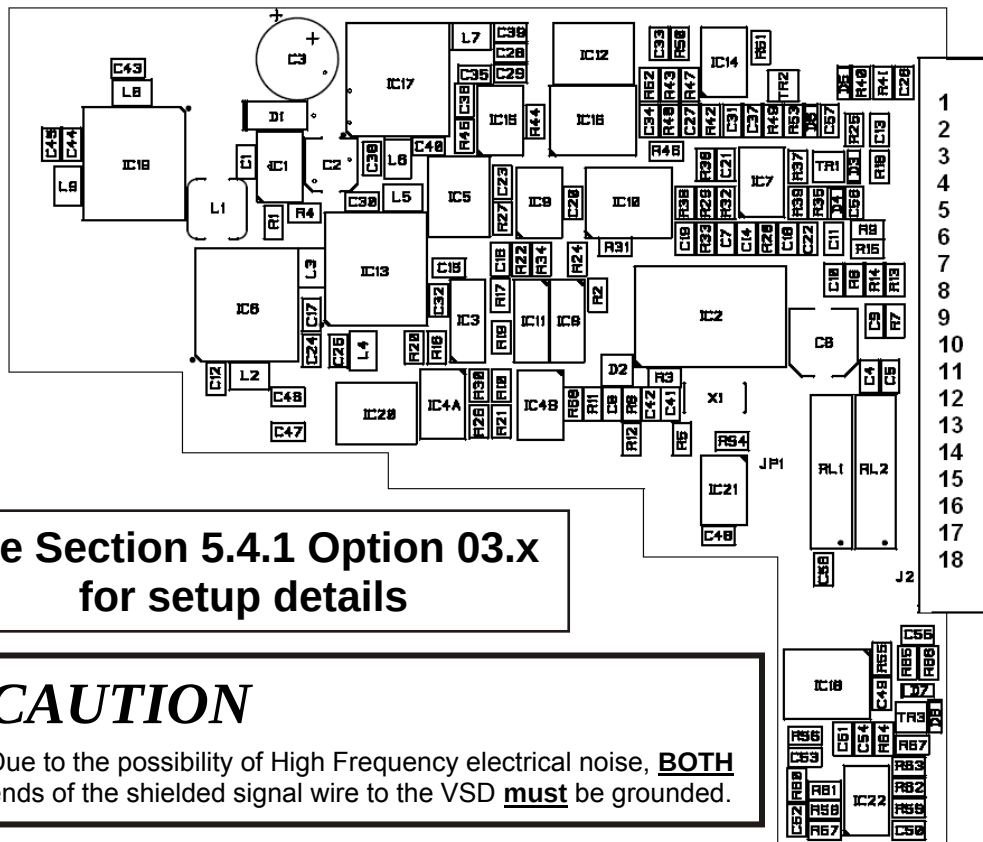
The NX6100 may have an optional daughter board within the main enclosure to allow control of 1 or 2 inverter drives. Optionally the daughter board is also available to allow feedback to be taken directly from encoders measuring the speed of the relevant motor, fan or pump shaft. There is also an option to allow the cup speed of an additional rotary device, e.g. atomizing cup, to be monitored as an encoder input.

When not configured to control an inverter, the 4-20mA outputs may be configured to transmit system variables as current outputs (4-20mA) such as modulation rate, steam pressure, etc.

There are special wiring requirements for inverter feedback. See Caution in Section 2.6.20.

Important note: When selecting a VSD or variable frequency drive (VFD) consideration must be given to the size and weight of the blower wheel. While a “fan” application is by nature a variable torque load, the inertia (weight) of the blower wheel must be considered. Typical HVAC type drives do not speed up or slow down large blower wheels with enough control to operate within the tolerance of the PPC or NX series control. Therefore, a VSD (VFD) with some built in breaking capability, such as a constant torque drive, should be used. At a minimum, if the drive has a constant torque setting, it must be enabled to minimize faults due to drive hunting. Since VFD’s are expected to perform as if they are servomotors, similar positional errors are applied to VFD’s. See Section 7.

2.6.20 PPC6000 Variable Speed Drive Daughter Board P/N NXDBVSD



CAUTION

Due to the possibility of High Frequency electrical noise, **BOTH** ends of the shielded signal wire to the VSD **must** be grounded.

- 4-20mA Output Channel 1 – PZ 1
- 4-20mA Output Channel 1 + PZ 2
- 4-20mA Output Channel 2 – PZ 3
- 4-20mA Output Channel 2 + PZ 4
- 4-20mA Output Channel 3 – PZ 5
- 4-20mA Output Channel 3 + PZ 6
- Counter Input 1 PZ 7
- +12Vdc PZ 8
- Counter Input 2 PZ 9
- +12Vdc PZ 10
- Counter Input 3 PZ 11
- Input Channel 1 – PZ 12
- Input Channel 1 ± PZ 13
- Input Channel 2 + PZ 14
- Relay 7 common PZ 15
- Relay 7 normally open PZ 16
- Relay 8 common PZ 17
- Relay 8 normally open PZ 18

The cabling for all of these terminals must be 2-conductor (core) overall screened PVC insulated 24 AWG (7/0.2mm). Since this cable may be run in conduit with high voltage wiring, it's voltage rating must exceed the maximum voltage carried by any other cable connected to the control or run in the same conduit.

The counter inputs are for use with encoder(s) for speed feedback from the devices being controller by the inverter(s) connected to the 4-20mA output(s) on this daughter board.

The analog outputs are isolated but the analog inputs are not and therefore special precautions must be taken when using these inputs.

The inputs are for 4-20mA signals **must** be isolated as the second input connects 'on-top' of the first to ensure 'cross-talks' will be detected. Terminal PX13 alternates ± to be shared with PZ12 and PZ14. This "switching" detects a direct link between input 1 and input 2 which is not allowed.

The function of relay 1 & 2 can be set by option parameter, but please note the maximum rating of these relays is 200mA at 50V.

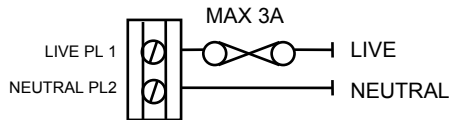
Incorrect connection may damage or destroy the units being connected.

2.6.21 LIVE and NEUTRAL supply (optional Oxygen Probe Interface unit)



WARNING

Incorrect setting of the Supply Voltage Links **WILL** damage or destroy the unit.



The LIVE and NEUTRAL supplies must be connected using multi-strand single conductor (core) PVC insulated 16 AWG (19/0.3mm) wire. The live connection should be fused with a maximum rating of 3A.

2.6.22 Oxygen probe connection (optional Oxygen Probe Interface unit), NX02INT

Bussman S504 series

PN GMA-5

5 AMP (Fast Acting)

(to protects transformer primary DC circuit)

Alternate Source:

Mouser Electronics : PN 504-GMA-5

PM 9- PROBE HEATER >24VAC
PM 8- PROBE HEATER >24VAC
PM 7- FLUE TEMP. THERMOCOUPLE +
PM 6- FLUE TEMP. THERMOCOUPLE -
PM 5- CELL TEMP. +
PM 4- CELL TEMP. -
PM 3- CELL OUTPUT +
PM 2- CELL OUTPUT -

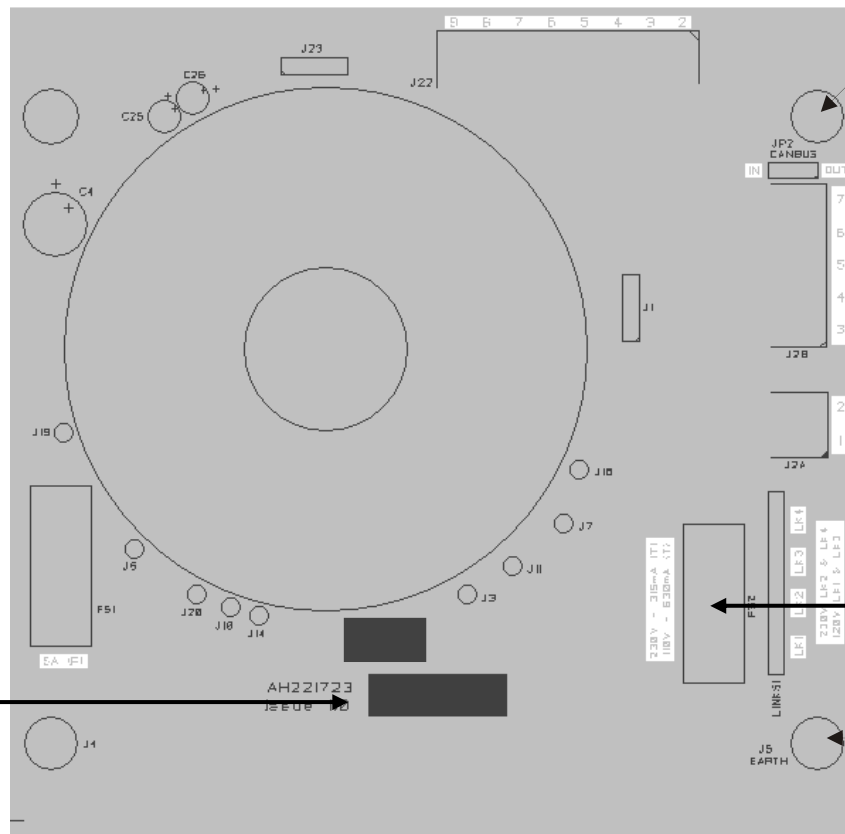
Bussmann - S504 series
P/N GMD-630 (TIME Delay) for
115V

P/N GMD-315 (TIME Delay) for
230V

(to protect 24V circuit)

Alternate Source::

Mouser Electronics



Termination for
screens from
Oxygen Probe (screw)

PL 7 0v Common (-) see note
PL 6 I/P 2 4 + 20mA (+) see note
PL 5 N/A future use
PL 4 CANbus -
PL 3 CANbus +

PL 2 Neutral
PL 1 Live

Ground
(Earth) (screw)

NOTE: 4-20mA O₂ input from non-Fireye O₂ probe. See sections 30.1 and 30.2



CAUTION

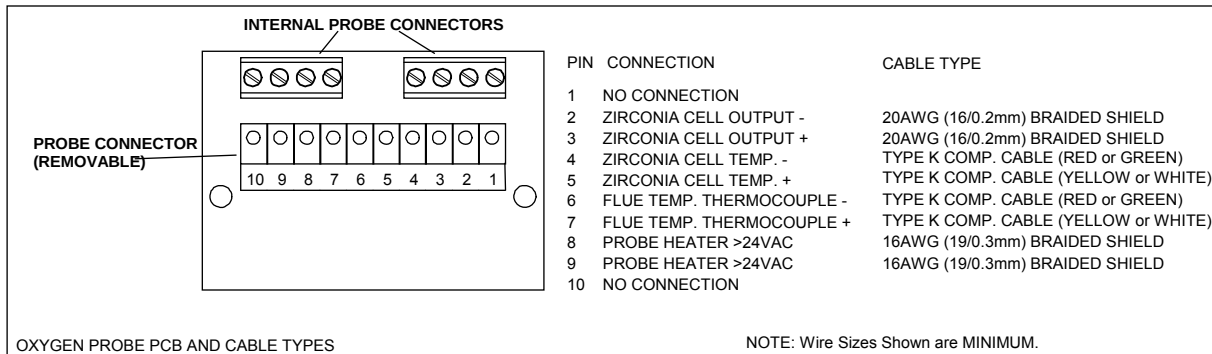
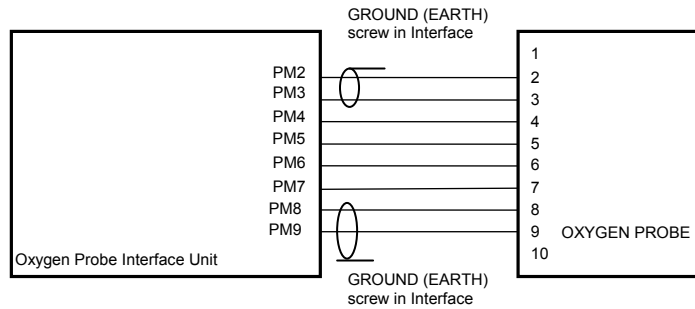
- Disconnect the power supply before beginning installation to prevent electrical shock, equipment and/or control damage. More than one power disconnect may be involved.
- Wiring must comply with all applicable codes, ordinances and regulations.
- Wiring where required, must comply with NEC Class 1 (Line Voltage) wiring.

The cabling between the unit and the probe consists of the following:

Cable	Specification
Probe heater cable	• Max. voltage in use <30V ac • Overall braid screened 2-core PVC insulated 16 AWG (19/0.3mm) (300V 176°F (80°C)). • Max. cable length 20 meters between units.
Probe cell signal cable	• Max. voltage in use < 15V dc • Overall braid screened 2-core PVC insulated 20AWG – 16/0.2mm (300V 176°F (80°C)). • Max. cable length 20 meters between units.
Heater and flue thermocouple compensation cable	• Max. voltage in use <15V dc • Cable Colors: ♦ EU -- 2 PVC insulated cores, green (+ve) and white (-ve) covered by a green PVC sheath. ♦ US -- 2 PVC insulated cores, yellow (+ve) and red (-ve) covered by a yellow PVC sheath. • Each core consists of 24AWG – 7/32AWG (300V 80°C) solid conductors covered by PVC insulation. (EU cable may be stranded wire.) • Type K compensating cable

NOTE: Thermocouple wires are polarized. Failure to connect the correct color to the designated terminal will result in a temperature offset that will affect the operation. CHECK TWICE to be certain.

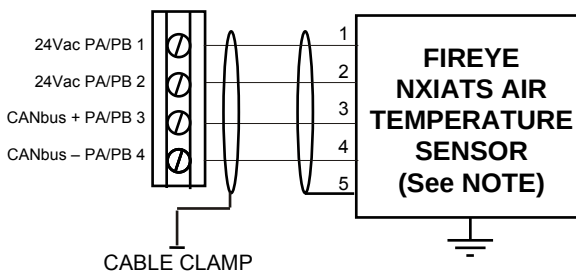
Oxygen Probe Wiring



After ensuring all connectors are correctly fitted, secure the probe cap on the probe. The heater supply must be connected using screened 2 conductor (core) 16 AWG (19/0.3mm) PVC insulated cable. The oxygen cell output must be connected using screened 2 conductor (core) 20 AWG (16/0.2mm) PVC insulated cable. Both thermocouple connections must be made using type K compensating cable. **For screen connection detail refer to section 2.6.2. DO NOT connect the screens at the probe.**

Incorrect connection may damage or destroy the units being connected.

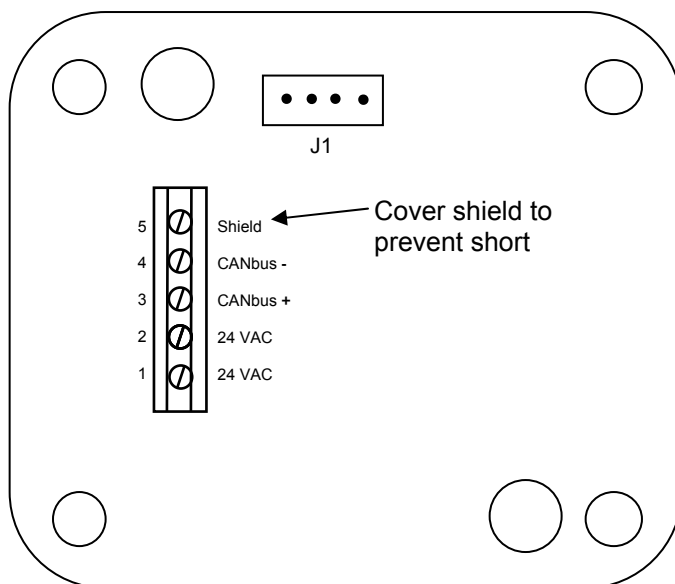
2.6.23 Inlet temperature sensor (optional)



NOTE: Insulate Shield (Terminal 5) to prevent accidental shorting to pins on circuit board (J1)

The system allows direct connection of an FIREYE inlet temperature sensor. The unit is connected via CANbus similar to the servo-motors and display, using overall screened 4 conductor (core) 24 AWG (7/0.2mm) PVC insulated cable. Since this cable is to be run adjacent to, and/or in the same conduit as high voltage wiring, its voltage rating must exceed the maximum voltage carried by any other cable connected to the control or run in the same conduit. **See Sections 2.6.1 & 2.6.2.**

Incorrect connection may damage or destroy the units being connected.



NOTE: Insulate Shield (Terminal 5) to prevent accidental shorting to pins on circuit board (J1)

NXIATS
Inlet Air Temperature Sensor

2.7 Final checks



WARNING

Incorrect setting of the Supply Voltage Links **WILL** damage or destroy the unit.

Before applying power, make the following checks:

- The control(s) supply voltage is correctly set, incorrect setting **will** destroy the unit(s).
- All other jumper (link) positions are correct.
- The fuse(s) are of the correct type and value, incorrect type or value may damage or destroy the units.
- All wiring and connections have been made according to the specifications detailed in this chapter.
- Before applying power to the control ensure that the probe heater output is not attempting to drive into a short circuit by checking between terminals PM8 and PM9 using a resistance meter. The expected resistance with an oxygen probe fitted should be greater than 15ohms.
- Enclosure lid(s) are fitted before applying power to the control(s).

3 Servo-motor selection and calibration



CAUTION

- All servo-motors connected to the PPC6000 control system **must** be approved as part of system.
- The total electrical load for all connected servo-motors **must not** exceed 40VA.
- Valve and motor alignment and calibration **must** be set in commission ratio mode prior to commissioning any profile setpoints.

3.1 Servo-motor selection

3.1.1 Motor requirements

Only servo-motors supplied by FIREYE may be used with this equipment, various servo-motors are available, to suit differing applications. The information below is intended as a general guide only; ensure that when installing or adjusting the servo-motors fitted to the appliance that reference is made to the correct instruction manual.

- All servo-motors must be connected via CANbus, no other possibility exists.
- All 24V servo-motors operate at 24Vac and are supplied from the PPC6000 (main) unit.
- All have a 90° maximum movement with an operating time of approximately 30 seconds for 90° travel.
- All have internal limit switches, which must be adjusted during commissioning to allow the close position check to be performed and to protect the burner/boiler against damage should a failure of the electronics occur. **NOTE: Any profile must NOT exceed 40VA.**

PPC6000 Servomotor Guide

Model	Rating FT/LB (Nm)	Voltage	VA
NXC04	3.0 (4Nm)	24	3
NXC12	8.75 (12Nm)	24	5
NXC20	15.0 (20Nm)	24	10
NXC40A	29.0 (40Nm)	24	18

Table: 3.1.1-A

Example: The profile requires 1 NXC20, 1 NXC40, 2 NXC04

Motor	VA
NXC04	6
NXC20	10
NXC40	18
Total	34

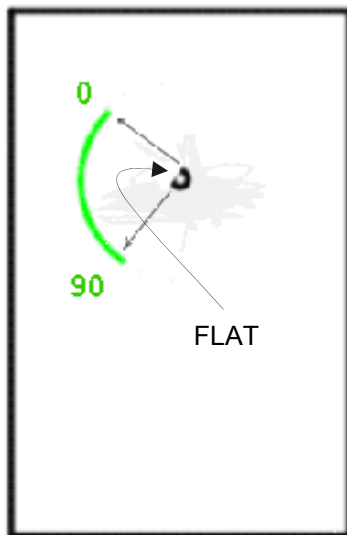
3.2 IMPORTANT: Servo-motor direction *warning*

It is necessary to ensure that each motor travels in the correct direction to match the mechanical configuration of the burner. The default setting is for a servo to move counter-clockwise (looking at the output shaft) as the motor is driven 'UP'. Option parameters 5.0 to 5.9 can be used to reverse this.

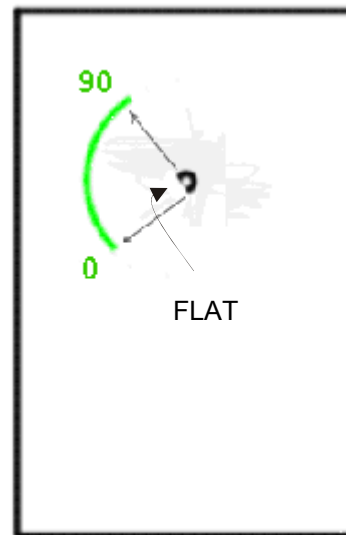
Servo-motors are shipped with the shaft flat in the 9-12 o'clock (facing motor shaft) position and by default rotate counter (anti) clockwise. For applications requiring clockwise rotation the motor must be left disconnected from the device being driven until after the system is powered and the motor set for clockwise direction via **Option Parameter 5.x**. Setting this option to "1" causes the motor to display the maximum travel degree (e.g. 90.0) as set by the internal adjustable end switch. After selecting all motors that are to travel in a clockwise direction, switch to the "adjust motors" display by pressing the "Mode" key. It is now possible to manually drive each of the clockwise motors to the new "P0" position (e.g. 1.0) as set by the adjustable end switch. The flat (facing the shaft) will now be in the 6-9 O'clock position. At this point the mechanical connection to the driven device can be made up. Fireeye "D" hole couplings should be used in order to assure a secure connection. Set screws alone are not recommended as they can come loose resulting in a potentially unsafe condition. Motor shafts are hardened generally preventing set screws from "biting" in and thus loosening after commissioning. Fireeye bulletin NEXBK-1000 provides information on "D" hole couplings. **Setting the option with the motor connected to the device being driven may cause damage to the device or motor.**

Note: The motor shaft, when directly coupled, will rotate in the opposite direction from the driven shaft. For example, if the gas valve rotates counter clockwise to open, then the servomotor **MUST** be set to clockwise rotation in Option 5.x

Shaft Positions as viewed from Shaft End
For NXC04, NXC12, NXC20, NXC40 ONLY



P0
Option Parameter 5.x
Set to 0



P0
Option Parameter 5.x
Set to 1

Servomotor Shaft Position Warning:

It is not possible to reposition the shaft of the servomotor other than to the factory settings of either clockwise or counter clockwise. That is, the shaft flat (as viewed from the shaft end) will either be in the 9-12 o'clock or 6-9 o'clock position at PO (burner off). Using the "out of the box" settings will make future replacement of servomotors easier. See previous page.

Some motors have a clutch release to allow for the driven device to be rotated for inspection after commissioning. It must be noted that releasing the clutch while the motor is powered will cause the servomotor to drive back to the commissioned P0 position. Power should be removed if the clutch is used for this purpose.

3.3 Locking the servo-motor to the valve shaft



WARNING

- Once the position of the servo-motor shaft relative to the valve shaft is correct, each motor **must** be locked to its valve shaft, for all air damper(s) and fuel valve(s).
- In order to prevent the joint between motor and shaft moving, it is recommended that the device used to link the two items is pinned together.
- It is essential that the motor and valve remain securely connected to ensure safe operation of the equipment, since feedback is only taken from the servomotor.
- Servomotor shafts are hardened; set screws alone are **NOT** sufficient to guard against slippage.

A variety of "D" hole couplings are available. Contact your Fireye Distributor, or Fireye, Inc., for assistance.

3.4 Feedback potentiometer



WARNING

- The servo-motor internal feedback potentiometer must be secured within the servo-motor to ensure that there is no possibility that the potentiometer can become disconnected from the motor output shaft.
- For direct drive servo-motor potentiometers (NXC40 only) ensure the potentiometer body is secured to the flexible mounting to prevent any movement between output shaft and potentiometer. Do not adjust the 'nuts' retaining the flexible mounting to the metalwork.
- For gear driven servo-motor potentiometers (4Nm, NXC04, NXc12, NXC20 servo-motor) ensure the potentiometer is secured to its mounting bracket and that the bracket is correctly attached to the servo-motor body to prevent any backlash between the potentiometer and the output shaft. **DO NOT ADJUST!**